

GROUP 5**AI Certification Exam Preparation & Assessment Assistant**

Track	Generative AI - LLM
Duration	2 Weeks (10 working days, 4 hours/day)
Dataset	No traditional ML training dataset required
API	Instructor-approved LLM API required
User Interface	Required

Project Analysis

Learners preparing for AI certification examinations may not know what to study, their current knowledge level, weak areas, or whether they are ready for the exam. This project builds an AI-powered assessment assistant that generates practice assessments and personalized feedback.

The system should evaluate learner answers, calculate results, identify weak areas, provide targeted recommendations, and create a study plan.

What the Student Must Build (Mandatory)

- Allow users to select a certification exam.
- Allow selection of difficulty level and number of questions.
- Generate a personalized practice assessment.
- Accept and evaluate learner answers.
- Calculate and display the final score.
- Show correct and incorrect answers.
- Identify weak knowledge areas.
- Explain incorrect answers.
- Provide targeted recommendations for improvement.
- Recommend topics for further study.
- Generate a personalized study plan.
- Display readiness level: Ready, Nearly Ready, or Needs Improvement.
- Include Home, Assessment, Result, and Feedback views in a functional UI.

Technical Requirements

- Python
- Approved LLM API
- Prompt Engineering
- Input Validation
- API error handling
- Environment variables for API keys
- Modular project structure

- Git & GitHub

Knowledge / Content Requirements

- Use reliable certification learning resources as the basis for topics and study guidance.
- Students should not claim that generated questions are official exam questions unless they come from an authorized official source.
- Test generated assessments for clarity, answer consistency, and useful feedback.

Optional Features (Bonus)

These features may improve the project but are not required for basic project completion.

- Assessment history
- PDF result report
- Email report
- Voice questions/answers
- Multi-language support
- Progress analytics dashboard
- Instructor/Admin dashboard
- Question bookmarking
- Adaptive difficulty

References and Starting Resources

- Microsoft AI-900 Learning Path
- Google AI Essentials
- AWS AI Practitioner
- IBM SkillsBuild AI
- DeepLearning.AI Short Courses
- Approved LLM API documentation